Abdallah Abdelhafeez Saber

Prospective Teaching Assistant / Demonstrator in Computer Science & Information Technology

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Academic Profile & Objective

High-achieving Computer Science and Information Technology graduate (**Cumulative GPA: 3.553 / 4.00, Very Good with Honors**; **Senior Graduation Project Grade: Excellent / Distinction**) and proven **Team Leader & Founder** across university capstones and regional hackathons. Demonstrates strong command of computer programming, database architectures, artificial intelligence, computer vision, and software engineering. Passionate about university pedagogy, laboratory instruction, student mentorship, and academic course development. Dedicated to pursuing postgraduate studies (M.Sc.) in Intelligent Systems, Generative AI, and Applied Machine Learning while actively supporting departmental teaching excellence and academic research.

Education & Academic Standing

Sinai University, Faculty of Information Technology and Computer Science **2022 – 2026**

Bachelor of Information Technology and Computer Science (Kantara Branch, Egypt) **Cumulative GPA: 3.553 / 4.00 (Very Good with Honors)**

- **Senior Graduation Project:** *RefereeX – AI-Powered Football Officiating Platform* | **Role: Founder & Team Leader** | **Grade: Excellent (Distinction / A+)**. Led the project team to build an automated decision-support pipeline combining YOLO, Segment Anything Model 2 (SAM 2), and Google Gemini 2.0 Flash Multimodal VLM for real-time video analytics.
- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming (C++, Java), Database Management Systems and SQL, Operating Systems, Computer Networks (Cisco CCNA curriculum), Software Engineering, Artificial Intelligence, Machine Learning, Deep Learning, Probability and Statistics, Discrete Mathematics, Web Technologies, Computer Architecture.
- **Academic Distinction:** Maintained consistent high academic standing on the Faculty Honor Roll across all four academic years.

Teaching Competencies & Laboratory Modules

1. Programming & Algorithms
CS101 Intro to Programming (Python, C++), OOP (C++, Java), Data Structures, Algorithms, Problem Solving.

2. Database Systems & SQL
Relational Database Management (RDBMS), SQL Querying and Optimization, Normalization, MS SQL Server, PostgreSQL.

3. AI & Machine Learning
Introduction to AI, Supervised and Unsupervised Learning, Deep Learning, Computer Vision (OpenCV), Prompt Engineering.

4. Data Science & Statistics
Data Wrangling, Exploratory Data Analysis (Pandas, NumPy), Statistical Computing in R, Power BI, Advanced Excel.

5. Computer Networks & IT Systems
Network Fundamentals (TCP/IP, Routing, Switching, VLANs), Cisco Packet Tracer, Windows and Linux System Administration.

6. Software Engineering & Web Tech
Web Fundamentals (HTML5, CSS3, JavaScript), Full-Stack Systems (PHP, Flask, FastAPI), Git and GitHub Version Control.

Research Interests & Focus Areas

- **Generative AI & Multimodal LLMs:** Retrieval-Augmented Generation (RAG) architectures, dense embeddings, vector databases (ChromaDB), hallucination reduction, and explainable AI in healthcare and decision support.
- **Applied Computer Vision & Video Analytics:** Real-time object detection (YOLO), promptable instance segmentation (Segment Anything Model 2 - SAM 2), and vision-language integration for automated spatial reasoning.
- **Predictive Analytics & Model Benchmarking:** Comparative algorithmic evaluation, ensemble techniques, high-dimensional feature selection, and data-driven healthcare predictive modeling.

Academic & Research Projects

RefereeX — AI-Powered Football Officiating Platform [Founder & Team Leader | Grade: Excellent] **2026 | GitHub**

Competition: Canal Startup Sprint Hackathon | Technologies: Python, Flask, YOLO, SAM 2, Gemini 2.0 Flash VLM, OpenCV

- Founded and led the project team developing an AI sports analytics venture, participating in the **Canal Startup Sprint Hackathon** and earning an **Excellent (Distinction / A+)** evaluation grade from the university examination committee.
- Managed the multi-object tracking pipeline using YOLO for player and ball detection, and integrated Segment Anything Model 2 (SAM 2) for zero-shot player segmentation and jersey-color team clustering.
- Integrated Google Gemini 2.0 Flash multimodal vision-language model to evaluate complex spatio-temporal game rules (fouls, offside lines, ball-out-of-bounds) with explainable natural language rationales.

LungGuide AI — Evidence-Grounded Medical Screening Assistant [Team Leader & AI Lead] **2026 | GitHub**

Competition: CREATIVA AI Hackathon | Technologies: LangChain, Python, FastAPI, React, TypeScript, ChromaDB, Gemini

- Led a team of student developers during the 40-hour intensive hackathon to architect an evidence-grounded Retrieval-Augmented Generation (RAG) assistant for lung cancer screening guidelines, delivering source-cited answers to minimize hallucinations.
- Engineered document ingestion, recursive text splitting, dense embedding generation, ChromaDB vector indexing, semantic search, evidence relevance scoring, and safety guardrails for non-grounded queries.

Academic & Research Projects (Continued)

Deep Learning Traffic Sign Recognition System [Lead Developer] **2026 | GitHub**

Technologies: Python, TensorFlow, Keras, CNN, OpenCV, Matplotlib, Google Colab

- Designed, trained, and evaluated a deep Convolutional Neural Network (CNN) architecture for multi-class traffic sign image classification.
- Implemented image normalization, histogram equalization, data augmentation, dropout regularization, and batch normalization, achieving high classification accuracy and robustness across unseen test splits.

Comparative Predictive Modeling & Customer Churn Analysis [Lead Analyst] **2025 | GitHub**

Technologies: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

- Conducted rigorous empirical benchmarking across classification models: Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, and Support Vector Machines to predict customer attrition.
- Handled class imbalance, performed hyperparameter tuning via GridSearchCV, and evaluated models using Confusion Matrices, ROC-AUC curves, Precision-Recall, and Gini Feature Importance.

Enterprise Relational Database Architecture & Query Optimization (SalesDB) [Database Architect] **2025 | GitHub**

Technologies: Microsoft SQL Server (SSMS), T-SQL, Relational Schema Design, Execution Plan Profiling

- Architected a normalized multi-table relational schema enforcing entity integrity, foreign key constraints, and transactional consistency; authored stored procedures, triggers, views, and optimized query execution plans through strategic indexing.

SkillUp — Web-Based Dynamic Learning Management System [Founder & Full-Stack Developer] **2025 | GitHub**

Technologies: PHP, JavaScript, SQL, HTML5, CSS3, Responsive Web Architecture

- Founded and developed an e-learning platform featuring role-based authentication (instructors and students), modular syllabus management, progress tracking dashboards, and automated certificate generation upon course completion.

Fellowships & National Initiatives

National Telecommunication Institute (NTI) — Digital Egypt Youth Program **Aug 2026 – Sep 2026**

IT Technical Support Specialist Track | Student ID: 241137 **Graduation Score: 99% (Distinction with Highest Honors)**

- Completed 120 intensive instructional hours (90 technical hours and 30 professional soft skills hours) covering computer hardware architectures, network protocols (TCP/IP, DNS, DHCP), Windows system administration, troubleshooting, and lab management.

Digital Egypt Pioneers Program (DEPI / MCIT) **Jun 2025 – Dec 2025**

Data Analyst Specialist Track — Ministry of Communications and Information Technology

- Completed advanced capacity-building coursework in end-to-end data pipelines, ETL processes, Power BI dashboard modeling, DAX measures, relational SQL querying, and business analytics.

Orange Digital Center, CREATIVA, ITIDA & TIEC AI Hackathon **Aug 2025 / 2026**

Intensive AI Training & Hackathon Accelerator | Role: Team Leader **Evaluation Score: 81.5%**

- Participated in an intensive 40-hour AI residency developing healthcare AI prototypes; led an engineering team to design and pitch LungGuide AI under faculty and industry mentorship.

Certifications & Specialized Courses

Building LLM Applications with Prompt Engineering — 2026
NVIDIA Deep Learning Institute (DLI)

AI for All: From Basics to GenAI Practice — 2026
NVIDIA Academy

Cisco CCNAv7 (Networking Fundamentals) — 2024
Cisco Networking Academy

Python for Data Science (Cognitive AI) — 2025
IBM Cognitive Class

SQL (Intermediate) & SQL (Basic) Verified — 2025
HackerRank

Intermediate SQL & Introduction to R — 2025–2026
DataCamp

Python Programming Basics — 2025
ITI Mahara-Tech (MCIT)

Data Science, Analytics & AI for Beginners — 2025
HP LIFE Academy

Technical & Pedagogical Skills

Programming & Scripting	Python, C++, Java, R, SQL (T-SQL, PostgreSQL, MySQL), PHP, JavaScript, HTML5, CSS3, Shell/Bash
AI & Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, OpenCV, YOLO, SAM 2, Hugging Face, Gemini API, RAG, ChromaDB
Data & Analytics	Pandas, NumPy, Matplotlib, Seaborn, Power BI, DAX, Power Query, Advanced Excel (PivotTables, Lookups)
Databases & Systems	Microsoft SQL Server (SSMS), PostgreSQL, MySQL, Cisco Packet Tracer, Windows Administration, Linux Environment
Leadership & Pedagogy	Technical Team Leadership, Undergraduate Lab Instruction, Problem Set Preparation, Code Review, Student Mentorship

Honors, Leadership & Languages

- **Academic Honors:** Graduated with 99% score from NTI Digital Egypt Youth Program; Top 5 Finalist in the National Data Challenge (500+ participants nationwide); 4-year Faculty Honor Roll at Sinai University (Cumulative GPA: 3.553 / 4.00).
- **Leadership & Entrepreneurship:** Founder and Team Leader for RefereeX, directing AI engineering and competing in the Canal Startup Sprint Hackathon; Team Leader for LungGuide AI at the CREATIVA Hackathon; led university study and lab groups.
- **Languages: Arabic** (Native); **English** (Upper-Intermediate / Professional Academic Proficiency for lecturing, technical lab instructions, and scientific research); **French** (Elementary / A1).
- **Academic References:** Available from faculty professors and department supervisors at the Faculty of Information Technology and Computer Science, Sinai University upon request.